

Independent Reading Report:

Language-Grounded Planning and Learning-Based Manipulation for Embodied Robots

Xingzhi Qian

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1 Reading Motivation

This reading project aims to understand how recent advances in language models, vision-language models, and generative policy learning can be connected with robot manipulation. Instead of treating embodied AI as a purely high-level reasoning problem, I focus on how semantic understanding can be grounded into executable robot actions, including task planning, functional grasping, and bimanual manipulation.

The five selected papers form a coherent technical path. The first paper, SayCan, studies how language models can be grounded through robot affordances. The second paper extends this idea to open-world task planning for humanoid bimanual dexterous manipulation. The third paper studies how language can guide functional grasping and hand synergy. The fourth paper introduces Diffusion Policy as a general framework for visuomotor policy learning. The fifth paper further connects diffusion-based policy learning with manipulability-aware bimanual manipulation.

My own background is in embedded robotics, state estimation, sensor fusion, real-time control, multi-agent decision-making, and robot learning. In the Fast Robots project at Cornell University, I worked with a small autonomous robot using IMU and ToF sensing, Kalman-filter-based distance estimation, yaw estimation, PID control, BLE-based logging, mapping, localization, and autonomous maneuver execution. In the Robot Learning project, I explored cooperative long-horizon task planning and embodied decision-making in Overcooked-AI.

These experiences motivated me to study embodied AI from a system-level perspective. A reliable robot should not only understand language or generate actions in simulation. It should connect perception, task reasoning, grasping, motion generation, and control into an executable physical pipeline. This reading project helps me connect my previous hands-on robotics experience with broader research questions in language-grounded robot planning and learning-based manipulation.

No.	Paper	Main Topic	Connection to My Background
1	Do As I Can, Not As I Say: Grounding Language in Robotic Affordances	Language grounding through robot affordances	Task decomposition, embodied decision-making
2	Open-World Task Planning for Humanoid Bimanual Dexterous Manipulation via Vision-Language Models	VLM-based humanoid task planning	Robot learning, long-horizon planning
3	Language-Guided Dexterous Functional Grasping by LLM Generated Grasp Functionality and Synergy for Humanoid Manipulation	Language-guided functional grasping	Command grounding, object manipulation
4	Diffusion Policy: Visuomotor Policy Learning via Action Diffusion	Generative visuomotor policy learning	Robot learning, action generation
5	ManiDP: Manipulability-Aware Diffusion Policy for Posture-Dependent Bimanual Manipulation	Physically grounded bimanual diffusion policy	Control feasibility, manipulation constraints

2 Selected Papers

3 Paper 1: Do As I Can, Not As I Say: Grounding Language in Robotic Affordances

Core Idea

The first paper studies how large language models can be grounded in real robot capabilities. A language model may generate a reasonable sequence of steps for a high-level instruction, but these steps are not always executable by a specific robot in a specific environment. SayCan addresses this problem by combining two types of information: the usefulness of an action for completing the instruction and the feasibility of that action for the robot.

The core idea can be summarized as

$$a^* = \arg \max_a p_{\text{LM}}(a \mid q, s) \cdot p_{\text{affordance}}(a \mid s),$$

where q is the language instruction, s is the current scene or robot state, p_{LM} measures how appropriate an action is according to the language model, and $p_{\text{affordance}}$ measures whether the robot can actually execute that action. The selected action must therefore be both semantically useful and physically feasible.

This formulation is important because it avoids using a language model as an unconstrained robot planner. Instead, the language model provides high-level semantic reasoning, while robot affordance models provide grounding in physical capability. This creates a more reliable bridge between natural language instruction and embodied action.

Takeaway

My main takeaway is that embodied language understanding should be constrained by robot capability. In robotics, a correct-looking plan is not enough; the plan must be executable by the robot. This idea connects to my Robot Learning project, where long-horizon tasks in Overcooked-AI also required decomposing high-level goals into feasible low-level actions. Although SayCan studies real robots rather than simulated agents, the underlying problem is similar: high-level reasoning must be grounded into what the agent can actually do.

This paper provides a useful starting point for understanding embodied AI. It shows that language models are powerful for semantic reasoning, but they need to be integrated with perception, affordance estimation, and control. This motivates the following papers, which move from general language grounding toward humanoid task planning and manipulation.

4 Paper 2: Open-World Task Planning for Humanoid Bimanual Dexterous Manipulation via Vision-Language Models

Core Idea

The second paper studies open-world task planning for humanoid bimanual dexterous manipulation. Compared with simple mobile robots or single-arm manipulators, humanoid robots face a more complex planning problem. They need to interpret visual scenes and language instructions, identify relevant objects, decompose tasks into executable steps, and coordinate two hands or arms during manipulation.

The paper uses vision-language models to support task planning in open-world environments. A simplified formulation is

$$\tau = \Pi_{\text{VLM}}(I, q),$$

where I is the visual observation, q is the language instruction, and τ is the generated task plan. The VLM helps connect visual context with task-level semantic reasoning. However, the generated plan still needs to be grounded into executable robot skills, such as reaching, grasping, transferring, placing, or bimanual coordination.

This paper extends the language-grounding idea from SayCan to a more complex humanoid setting. The challenge is no longer only whether a robot can execute a skill, but also whether the plan is compatible with dexterous manipulation, object affordances, and bimanual execution.

Takeaway

My main takeaway is that open-world humanoid manipulation requires both semantic reasoning and physical grounding. A VLM can help the robot understand the task context, but task completion still depends on low-level manipulation skills and reliable execution.

This connects to my Overcooked-AI project because both involve long-horizon embodied task planning. In Overcooked-AI, agents must complete a sequence of actions such as collecting ingredients, cooking, serving, and delivering. In humanoid manipulation, the action space is much richer and more physically constrained. This paper helped me understand how task planning becomes more challenging when the agent is a real embodied robot with hands, arms, posture, and object interaction constraints.

5 Paper 3: Language-Guided Dexterous Functional Grasping by LLM Generated Grasp Functionality and Synergy for Humanoid Manipulation

Core Idea

The third paper focuses on language-guided dexterous functional grasping. Traditional robotic grasping often emphasizes geometric stability, but human-like grasping is strongly task-dependent. The same object may need to be grasped differently depending on whether the robot wants to use it, pass it, pour from it, or place it somewhere.

The paper uses large language models to infer grasp functionality and hand synergy. A simplified representation is

$$(f, s) = \Phi_{\text{LLM}}(o, q),$$

where o is the object description, q is the task instruction, f is the inferred grasp functionality, and s is the corresponding hand synergy. The grasp functionality describes the purpose of the grasp, while hand synergy reduces the complexity of dexterous hand control by representing coordinated finger movements through lower-dimensional patterns.

The key contribution is that language is not only used for high-level task planning. It also influences the physical form of manipulation. This is important for humanoid robots because dexterous hands have many degrees of freedom, and the correct grasp depends on both object geometry and task intention.

Takeaway

My main takeaway is that manipulation should be understood as task-conditioned physical interaction. A stable grasp is not necessarily a good grasp if it does not support the intended downstream action. Language can provide useful semantic information about how an object should be used, but this information must still be converted into feasible hand configurations and control commands.

This paper connects to my interest in embodied AI because it shows how high-level semantic reasoning can affect low-level robot behavior. In my embedded AI project, voice commands were used to trigger device-level actions. This paper studies a more advanced version of the same general problem: how language can be grounded into physical interaction with the world.

6 Paper 4: Diffusion Policy: Visuomotor Policy Learning via Action Diffusion

Core Idea

The fourth paper introduces Diffusion Policy for visuomotor policy learning. Instead of predicting a single deterministic action, Diffusion Policy represents robot behavior as a conditional denoising diffusion process. Starting from noisy action sequences, the model gradually denoises them into executable actions conditioned on visual and state observations.

A simplified denoising formulation is

$$\mathbf{a}_{k-1} = f_{\theta}(\mathbf{a}_k, o, k),$$

where $\mathbf{a}_k$ is the noisy action sequence at denoising step k , o is the observation, and f_θ is the learned denoising model. After multiple denoising steps, the policy outputs an action sequence for robot execution.

This approach is powerful because robot manipulation often has multimodal action distributions. For the same task, there may be multiple valid ways to reach, grasp, or manipulate an object. A diffusion-based policy can better represent this diversity than a simple deterministic policy. The paper also connects diffusion policy with receding horizon control, which makes it more suitable for closed-loop robot execution.

Takeaway

My main takeaway is that generative models can be used not only for images or language, but also for robot action generation. Diffusion Policy provides a flexible framework for learning complex visuomotor behaviors from data.

This paper is important for my reading path because it provides the technical foundation for understanding more specialized diffusion-based manipulation methods. It also connects to my Robot Learning background. In embodied tasks, the agent often needs to choose actions under uncertainty and partial observation. Diffusion Policy suggests that modeling a distribution of possible actions can be more effective than predicting a single action directly.

7 Paper 5: ManiDP: Manipulability-Aware Diffusion Policy for Posture-Dependent Bimanual Manipulation

Core Idea

The fifth paper builds on diffusion-based robot policy learning and applies it to posture-dependent bimanual manipulation. In bimanual manipulation, generating an end-effector trajectory is not enough. The robot must also consider whether its current posture allows the motion to be executed effectively. The same target motion may be easy in one configuration but difficult or unstable in another.

The paper incorporates manipulability into diffusion policy learning. A common manipulability measure is

$$w(q) = \sqrt{\det(J(q)J(q)^T)},$$

where q is the robot joint configuration and $J(q)$ is the Jacobian matrix. Higher manipulability generally means that the robot can move more flexibly in task space around the current posture.

The key idea of ManiDP is that a learned policy should not only imitate demonstrated actions, but also consider physical feasibility. By introducing manipulability awareness, the policy can generate bimanual actions that are more compatible with the robot's posture and kinematic structure. This makes the method more physically grounded than a purely data-driven action generator.

Takeaway

My main takeaway is that robot learning should be combined with classical robotics constraints. Diffusion policy provides expressive action generation, but manipulability gives the policy a physically meaningful criterion. This combination is especially important for humanoid or dual-arm systems, where the

action space is high-dimensional and strongly constrained by body posture.

This connects to my Fast Robots project because I learned that a control command is only useful if it can be executed reliably by the physical robot. Even in a small mobile robot, motor response, yaw estimation, sensor noise, and PID tuning directly affected performance. ManiDP shows a similar principle in a more advanced manipulation setting: learned actions must respect the robot’s physical capability.

8 Integrated Understanding Across the Five Papers

After reading these five papers, I understand embodied robot intelligence as a layered system that connects language, perception, planning, policy learning, and control.

SayCan shows that language-model reasoning should be grounded by robot affordances. The open-world humanoid task planning paper extends this idea to vision-language reasoning for bimanual manipulation. The language-guided functional grasping paper moves from task-level planning to object-level and hand-level manipulation. Diffusion Policy provides a general framework for learning visuomotor action generation. ManiDP further shows that learned policies should incorporate physical feasibility, such as manipulability and posture constraints.

Together, the five papers form the following technical path:

Language Grounding → Vision-Language Task Planning → Functional Grasping → Diffusion-Based Policy Learning

This path helped me understand that embodied AI is not simply about applying large models to robots. A robot must be able to ground language in perception, convert plans into object interactions, generate feasible actions, and execute them under physical constraints.

9 Connection to My Background and Future Direction

My Fast Robots project gave me hands-on experience with a complete robotic pipeline: sensing, filtering, control, logging, and behavior execution. Although the platform was much simpler than a humanoid robot, it taught me that reliable autonomy depends on the interaction between algorithms and physical constraints. This experience connects most directly to the physically grounded aspects of Diffusion Policy and ManiDP.

My Robot Learning project gave me experience with long-horizon embodied decision-making. In Overcooked-AI, agents needed to decompose a task into sub-goals, coordinate actions, and complete object-handling procedures. This connects to SayCan and the VLM-based humanoid task planning paper, both of which study how high-level goals can be transformed into executable action sequences.

My embedded AI and voice-command system experience also supports this reading direction. In a simple embedded system, natural language commands can trigger predefined actions. In embodied robot manipulation, however, language must be grounded into object affordances, grasp types, motion policies, and control constraints. This reading project helped me understand the deeper research challenge behind language-conditioned robot behavior.

A possible future direction is to study how foundation models and learning-based policies can be connected with lightweight real-robot platforms. For example, a small robot could use language or visual input for task-level reasoning, while local perception and control modules ensure safe execution. This could provide a practical bridge between my current embedded robotics background and more advanced

research in humanoid embodied intelligence.

10 Potential Research Questions

Based on these readings, I am interested in the following research questions:

1. How can vision-language models be grounded more reliably into executable robot manipulation actions?
2. How can language-level task reasoning be connected with low-level grasping and motion policies?
3. How can diffusion-based robot policies incorporate physical feasibility, such as manipulability and posture constraints?
4. How can lightweight real-robot platforms be used to test ideas in embodied planning, perception, and control before transferring them to more complex humanoid systems?

11 Conclusion

This reading project provided a structured understanding of language-grounded planning and learning-based manipulation for embodied robots. The five papers show that robot intelligence requires more than a language model, a planner, or a learned policy alone. It requires an integrated pipeline that can understand tasks, evaluate robot affordances, reason about object functionality, generate actions, and respect physical constraints.

For my future research preparation, the most important insight is that embodied AI should be treated as a grounding problem. Language and vision models provide high-level semantic reasoning, but this reasoning must be connected to robot skills, object interactions, and physical feasibility. Similarly, diffusion-based policies provide powerful action generation, but they must be constrained by the robot's kinematics and control requirements.

My background in embedded robotics, Kalman-filter-based estimation, PID control, multi-agent decision-making, and robot learning provides a practical foundation for this direction. Building on these readings, I am particularly interested in research that connects foundation models, robot learning, and control for real embodied systems.

References

- [1] M. Ahn, A. Brohan, N. Brown, Y. Chebotar, O. Cortes, B. David, C. Finn, K. Gopalakrishnan, K. Hausman, A. Herzog, D. Ho, J. Hsu, J. Ibarz, B. Ichter, A. Irpan, E. Jang, R. Julian, D. Kalashnikov, Y. Kuang, I. Leal, S. Levine, Y. Lu, U. Malla, D. Manjunath, I. Mordatch, O. Nachum, C. Parada, J. Peralta, E. Perez, K. Pertsch, J. Quiambao, K. Rao, M. Ryoo, G. Salazar, P. R. Sanketi, K. Sayed, J. Singh, S. Sontakke, A. Stone, C. Tan, H. Tran, V. Vanhoucke, S. Vega, Q. Vuong, F. Xia, T. Xiao, P. Xu, S. Xu, M. Yan, and A. Zeng, “Do As I Can, Not As I Say: Grounding Language in Robotic Affordances,” *Proceedings of the 6th Conference on Robot Learning*, pp. 287–318, 2023.
- [2] Z. Tang, Z. Li, J. Liu, Z. Li, and F. Chen, “Open-World Task Planning for Humanoid Bimanual Dexterous Manipulation via Vision-Language Models,” *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 21551–21558, 2025.

- [3] Z. Li, J. Liu, Z. Li, Z. Dong, T. Teng, Y. Ou, D. G. Caldwell, and F. Chen, “Language-Guided Dexterous Functional Grasping by LLM Generated Grasp Functionality and Synergy for Humanoid Manipulation,” *IEEE Transactions on Automation Science and Engineering*, vol. 22, pp. 10506–10519, 2025.
- [4] C. Chi, Z. Xu, S. Feng, E. Cousineau, Y. Du, B. Burchfiel, R. Tedrake, and S. Song, “Diffusion Policy: Visuomotor Policy Learning via Action Diffusion,” *The International Journal of Robotics Research*, vol. 44, no. 9, pp. 1281–1315, 2025.
- [5] Z. Li, J. Liu, D. Li, T. Teng, M. Li, S. Calinon, D. Caldwell, and F. Chen, “ManiDP: Manipulability-Aware Diffusion Policy for Posture-Dependent Bimanual Manipulation,” *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 9956–9962, 2025.